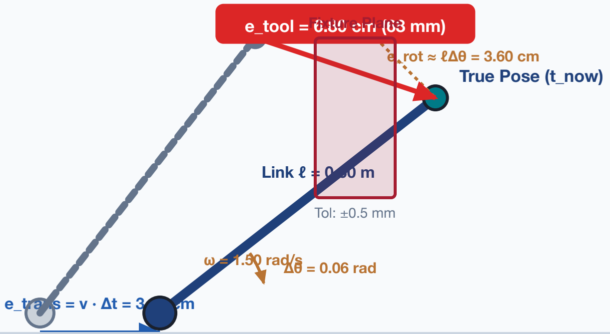
COORDINATE TRANSFORM STALENESS & LEVER-ARM ERROR PROPAGATION

Rigid-Body Kinematic Divergence: $e_{tool} \approx v \cdot \Delta t + \ell \cdot \omega \cdot \Delta t$ Under Stale Coordinate Frames

PHYSICAL ARM DYNAMICS & LEVER ARM DIVERGENCE



CRITICAL COLLISION: 68.0 mm Error vs 0.5 mm Mechanical Clearance (136× Breach)

Uncompensated 40 ms lag drives end-effector into solid fixture during trajectory execution.

KINEMATIC LEVER-ARM ARITHMETIC

- **Translational Drift:**
  $e_{trans} = v \cdot \Delta t = 0.80\ \text{m/s} \times 0.040\ \text{s} = 3.20\ \text{cm}$
- **Rotational Arc Deflection:**
  $e_{rot} \approx \ell \cdot (\omega \cdot \Delta t) = 0.60\ \text{m} \times (1.50 \times 0.040) = 3.60\ \text{cm}$
- **Total Tool-Point Offset:**
  $e_{tool} \approx e_{trans} + e_{rot} = 3.20 + 3.60 = 6.80\ \text{cm}$ (68.0 mm)
- **Task Safety Clearance:**
  $\varepsilon_{tol} = 0.50\ \text{mm}$ (Exceeded by 136× within 40 ms!)

LOUD VS SILENT TRANSFORM FAILURES

✓ LOUD FAILURE: MISSING / DROPPED TRANSFORM

- Sensor link drops; frame tree records null edge.
- Matrix traversal raises FRAME_NOT_FOUND exception.
- **Result: Deterministic emergency stop / safe position hold.**

× SILENT FAILURE: FROZEN / REPUBLISHED MATRIX

- Middleware republishes cached pose with fresh OS timestamp.
- Downstream linear algebra evaluates with zero exceptions.
- **Result: Trajectory planner drives tool into solid fixture.**

RULE: Transforms MUST carry immutable evidence epoch t_e.